

→ Pressman (S.E)
→ Agarwal (S.E)
K.K

Contents :-

- ① what is software ?
- ② characteristics of sw.
- ③ sw Appl's.
- ④ sw Engineering
- ⑤ Process models
- ⑥ Estimation models
- ⑦ Risk Analysis
- ⑧ SDLC

Software :-

→ sw is a prog. . when it is executed the associated functionality must be satisfied.

→ Prog. consist the d.s used to manipulate the info.

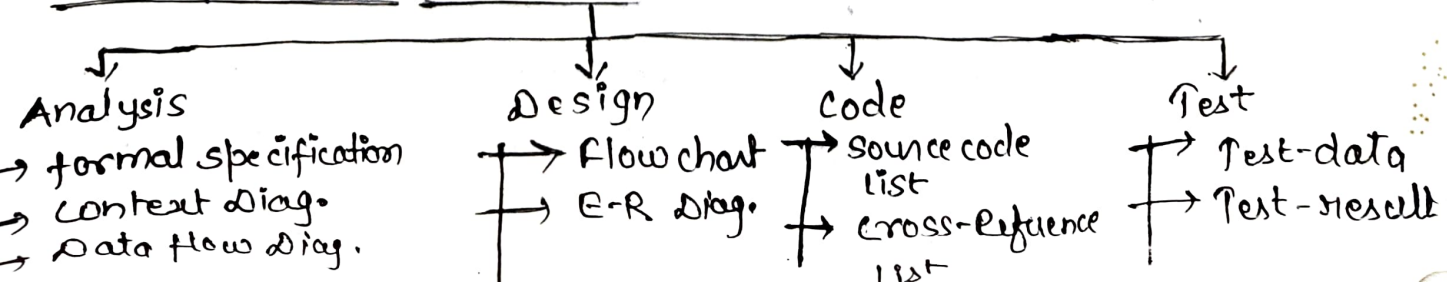
sw contain the operational procedures used to describe the instr's how to operate the system.

→ The operational procedures are :-

- User manuals
 - Syst. overview
 - Beginner's guide
 - Reference guide
- Operational manuals
 - Installation guide
 - Administration guide.

sw contained the documentation manuals used to describe the systematic Approach used to develop the product.

Documentation manuals are



→ Finally, we can conclude slw is a combination of prog., operational procedures & documentation manual.
So slw is a logical component rather than physical component.

② Characteristics of the slw :-

→ slw is developed/engineering but not manufactured in the classic sense.

Note → slw is not having the assembly line begg. prog. to prog. the functionality is diffu.

→ Only one time the prog. is developed that can be copied into multiple copies. During this process no need to start the development from the base line

→ whereas in the physical component development, assembly line is required. Each product requires separate raw material.

→ slw ~~is~~ does not "wear out" but "deteriorate".

→ physical component undergoes wear-out. The life cycle of the physical component contain 3 phases name as:-

① Burn-in phase

② useful life phase

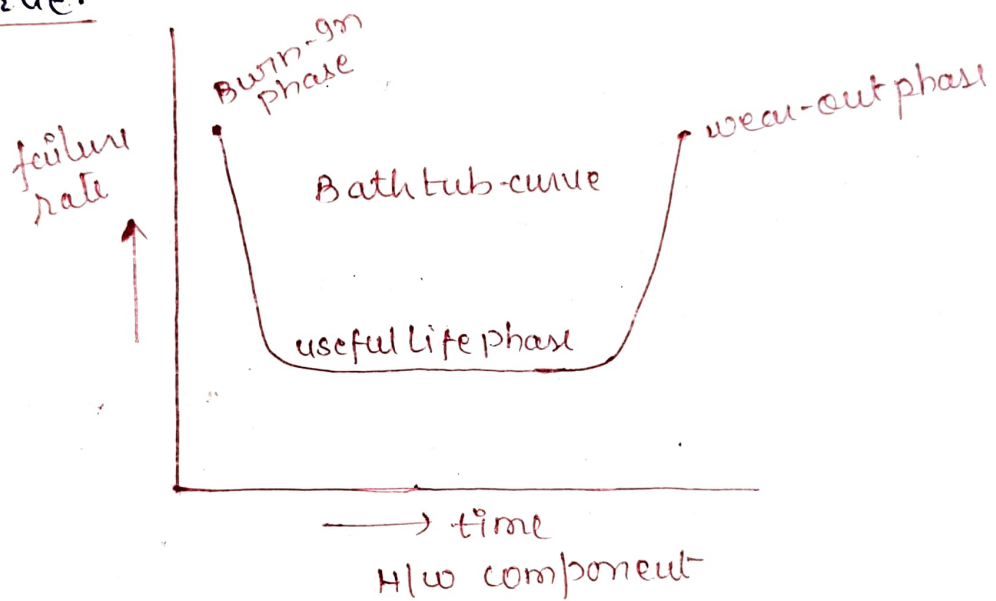
③ wear-out phase

→ In the early stage of development due to the more no. of errors the product failure rate is very high. This stage is called as Burn in phase.

→ After implementing the different test opns. the errors will be removed from the product. So At particular point the failure rate will be stabilized. So the product will be deployed into the customer side.

→ After Deployment the developed product is in the operational stage called as useful life phase.

- After continuous uses of the product over a period of time the life time of a product will be decreases due to the environmental changes (Temp., Dust etc.)
- So the failure rate of the product again increases. This state is called as wear-out phase.
- when the physical component undergoes wear-out then replace the component with new component.
wear-out condⁿ can be described by using the bath-tub curve.

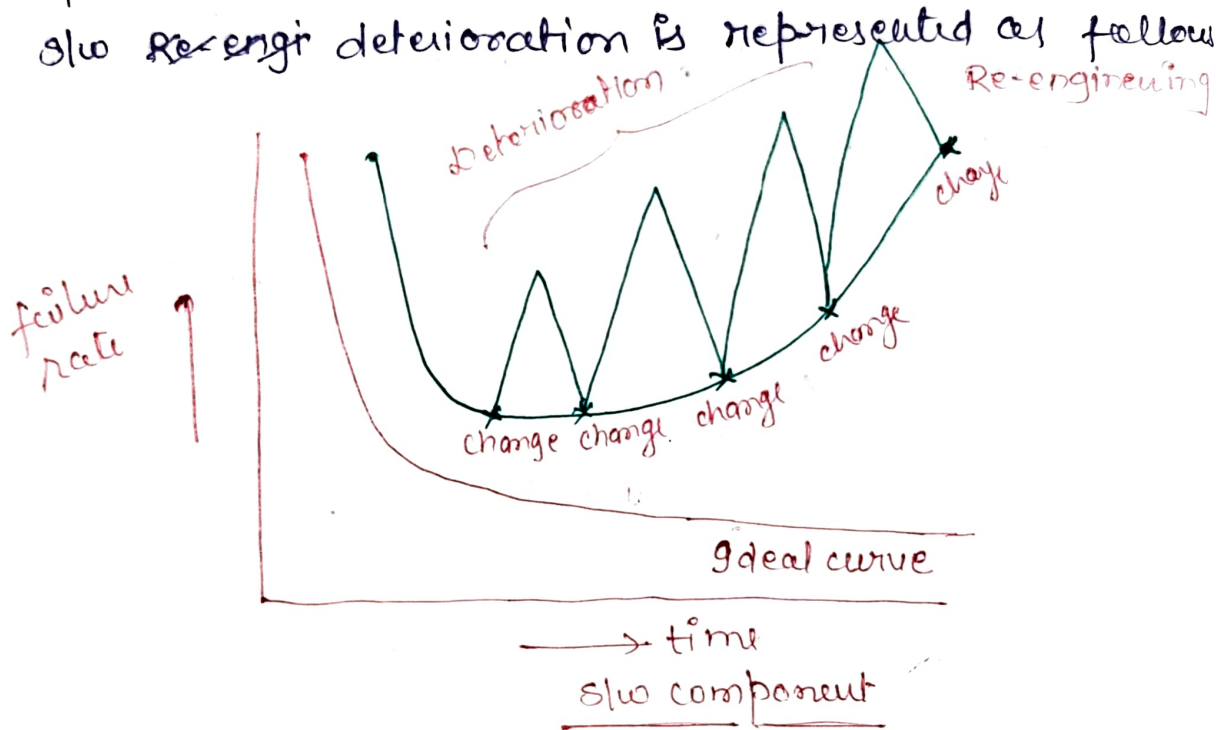


- Logical component is undergoes deterioration but it never undergoes wear-out
- In the early stage of development the product failure rate is high, after the testing the product failure rate will be stabilized.
So the product will be deployed into the operational state.
- when the logical component is operated without any changes the product-failure rate is never increases bcoz the logical component is never effected by the environmental changes.
So it never enter into the wear-out.

→ But the customer requirements are not static. So due to the frequent changes the product maintenance cost becomes twice or thrice than the development cost. This condⁿ is called as the deterioration.

→ when the slw is undergoes deterioration then Re-engineer the slw.

The slw Re-engi deterioration is represented as follows.



7th Nov

→ Industry is moving towards component based development still slw is a custom built.

- Component is a reusable code. It is also called as fully tested code or error free code or Risk free code.
- when the components are used in the project development the development cost & time will be minimized. So we can develop the project on-time & within the Budget.
- H/w components are directly used in the project without any change.
- But the slw components are not used in the project directly bcoz project to project the functionality is

diff. so some modification is required in the component. This is called as the customization.

→ In the SW development process 4 types of the components are used.

- ① Off-the-shelf component: These components are extracted from the 3rd party libraries.
- ② Fully Experienced } These components are extracted from
- ③ partial " } the own orgn. library.
- ④ New component: This component functionality is developed from the Base-line.

③ SW Applications :-

There are different appl's uses the SW to satisfy the various high-level objectives.

① System SW :- System SW is a prog. used to manage the other prog. ~~and~~ provides the services to the other programs
eg: compiler, editor, O.S, Drivers etc.

② Real-time SW :- Real time SW is used to monitor, analyze & control the Real-time events as they occur.

→ The SW developed in the weather-forecasting applⁿ is called as Real-time SW.

→ This SW is used to gather & process the status of temp, humidity & other environmental parameters to forecast the weather.

③ Embedded SW :- This SW is placed in the permanent mem of the product & it is used to control various functⁿ of the product.

eg: Aircraft, automobile, security system, signaling system etc.

→ Embedded sw is used to control the h/w component functionality so it is also called as the intelligent sw.

④ Business Applⁿ/sw :- This is a largest applⁿ area. Business sw is used to process the different business functⁿs.
eg: payroll, file-monitoring, Employee mgmt. system,
Account mgmt. system.

→ Data ware housing also comes under the business applⁿ domain.

→ Different ETL (Extraction, Transformation & Loading) tools are available to convert the raw data into meaningful info. with respect to business functⁿ.

eg: MIS (mgmt info. sys.)
ERP (Enterprise Resource planning)

⑤ Engineering & scientific sw :- This sw is used to perform the complex computation.

eg: MATLAB, NS2, CAD/CAM packages, Circuit Analysis etc.

⑥ Personal sw :- This sw is used in the personal computer.

eg: word-processing, Graphics, multimedia, Games etc.

⑦ web-sw :- It is used to develop the user interface.

→ It provides the different web-services for different web-applⁿs.

eg: CGI (Common Gateway Interface)
Java script
perl
HTML

- ③ Artificial Intelligence slw:- In this category, symbolic computation are perform rather than the numeric computation
- Functional programming languages are used to perform the symbolic computation.
 - Ex. of functional lang. LISP (LIST PROGRAMMING)
PROLOG (LOGIC ")
ML (meta lang.)
Haskell
 - Appl's are :: m/ch. Learning (Robot design)
Artificial Neural n/w^s
Expert system
Signal processing etc.

Conclusion To develop the efficient, quality slw in the different appl^y. domains to satisfy the various high-level objectives there is a need of the system approach. i.e slw engineering.

- ④ Slw engineering:- The appl^y of systematic, disciplined, & quantifiable approach to the development, opⁿ. & maintenance of slw called as slw engineering.

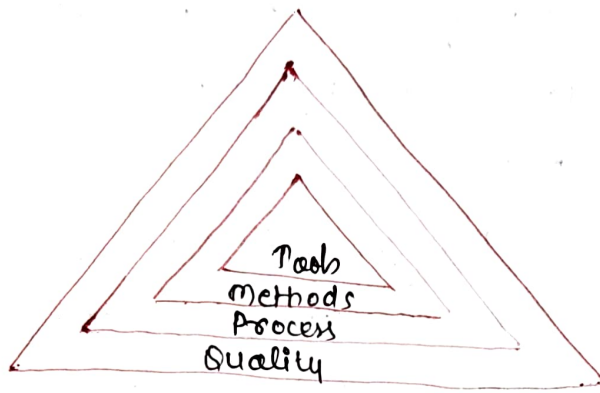
→ slw Eng. uses the two approaches to develop the reliable slw.

- ① Layered Approach
- ② Generic "

- ① Layered Approach:- This approach contains 4 layers to develop the quality product.

- | | |
|------------------|------------------|
| a) Quality layer | c) methods layer |
| b) Process " | d) Tools " |

→ the priority assignment of these 4 layers are shown below



a) Quality:- Quality layer is bedrock of the s/w development process to develop the quality product.

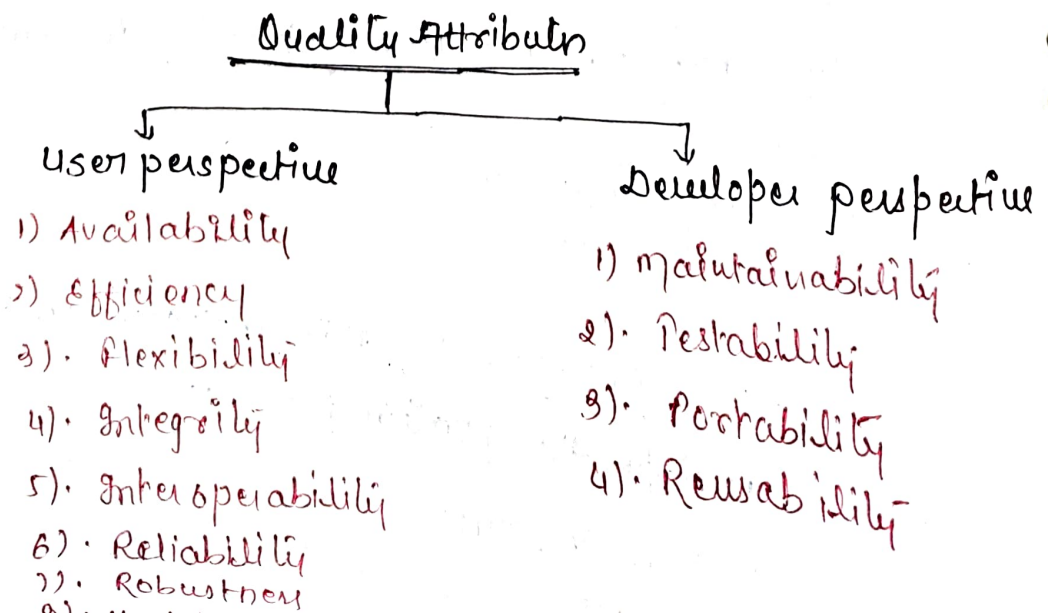
→ Acc. to IEEE standard "quality is defined as conformance to the explicitly stated requirements."

→ Quality is an indirect measurement, it is depending on the no. of the defects. the relationship b/w the quality & defects is

$$Q \propto \frac{1}{\text{Defects}}$$

→ Defect is an uncovered errors after careful evaluation of the project.

→ Diff. Quality attributes are present to define the quality factor. the classification of the quality attributes shown below:-



b). Process layer :- This layer is focused on the key-process area (KPA) to develop the quality product.

→ In the slow development 4 p's are involved :-

Priority
level
↓

4 p's : 1) People : efficient developers & managers

2) Product : what we are going to deliver to the customer. The deliverables may be a prog. or operational procedures or documentation manuals.

3) Process : The way which is used to develop the product is called as process.

4) Project : systematic estimated plan to develop the product.

③ Methods layer :- This layer is focused on the process models. process models ~~are~~ shows the framework to develop the product.

④ Tools :- This layer is focused on the automated & semi-automated tools available in the 4th generation Techniques. to select the appropriate background lang.

② Generic Approach :- Generic approach contain 3 stages to develop the sks.

grp.

a) Definition b) Development c) Support

a) Definition :- This stage is focused on "what" that means

→ what info. to be processed. ⇒ inflows (input)

→ what functionality & performance is desired? ⇒ Quality attribute

→ what behaviour is expected ⇒ out-flows (output)

→ what constraints are imposed ⇒ Limitation

→ what validation criteria is implemented ⇒ Right requirement

- In this stage requirements analysis activities performed.
- In the S.E validation & verification processes plan the plays the important role.

Validation:- "Are we build the right system"

- In this process the right condⁿ is injected at the early stage of SDLC.

Verification:- "Are we build the system right."

- This process is implemented after developing the prog. that means it is implemented during the testing.

b) Development:- This stage is focused on "How" that means

- How to define D.O.S
 - How to create architecture.
 - How to define interfaces
- } ⇒ Design
- How to translate procedural description into m/cⁿ, Readable format ⇒ coding
 - How to develop test cases & how to implement test-cases ⇒ Testing

- In this stage, design, coding & Testing activities are performed.
- After the testing the error-free logical component is deployed into the customer's site.

c) Support:- Support stage is focused on the "Maintenance"

- After deploying the product into the customer's place when it is in the operational stage support stage is required to maintain the product from

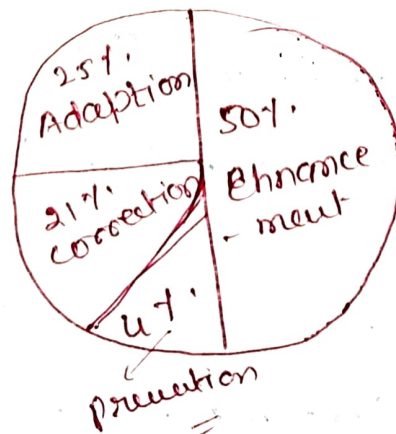
→ 4 types of the support required to maintain the product:-
Support Maintenance

- | | | | |
|----------------|---|------------|---------------|
| a) correction | → | Corrective | } <u>grp.</u> |
| b) Adaption | → | Adaptive | |
| c) Enhancement | → | Perfective | |
| d) prevention | → | Preventive | |

- corrective maintenance is required to maintain the product from the uncoured errors.
- Adaptive maintenance is required to maintain the product from the subsystem changes. (CPU/mem/O.S. upgradation)
- perfective maintenance is required to maintain the product from the functional requirement changes.
- preventive maintenance is required to maintain the product from the frequent changes.

→ the effort required to maintain the s/w is called as
Maintenance Effort

→ the maintenance effort distribution is shown - below



Priority

Enhancement	↓
Adaption	
Correction	
Prevention	

Assessment model :-

- To access the s/w orgn. based on the type of the process used to develop the product there is a need of the assessment model.
- Assessment model is required to improve the process maturity levels.
- s/w Eng. Institution define one assessment model name as CMM (Capability maturity model) ✓

CMM :-

→ This model consists 5 levels of the assessments name as

- ① Level 1 (Initial) → 0 KPA's
- ② Level 2 (Repeatable) → 6 KPA's
- ③ Level 3 (Defined) → 7 KPA's
- ④ Level 4 (Managed) → 2 KPA's
- ⑤ Level 5 (Optimized) → 3 KPA's

① Initial :- (Level 1) :

- In this level, "Adhoc" procedures are used to develop the product.
- ∴ The process becomes the "Chaotic" (Confusion)
- In this level success or failure is depending on the individual effort.

② Repeatable :- (Level 2)

- In this level project mng. activities are derive based on the experience of the previous project & estimate the current project.
- No guarantee to develop the project on time.

③ Defined :-

- project mgmt. & eng. activities are documented but not quantitatively controlled.
- so ~~no~~ soln. for the Anticipated situation.
(Sudden Surprise/Risk)

④ Managed :-

- project mgmt. & eng. activities are quantitatively & qualitatively at the managed level.
- But due to lack of continuous improvement process min. quality will be assured.

⑤ Optimized :-

- Gain more profit with min. effort by taking the continuous qualitatively feedback from the customer & by using the innovative ideas & tools.
- In this level continuous improvement process is present so the customer satisfaction is very high.

8th NOV

⑤ Process model :-

- Process is a framework.
- Framework indicates the sequence of the actions.
- In the s/w eng. practices different types of the frameworks are present to develop the different products.
- The classification of frameworks shown below.

